

Advanced OpenMP

Newer Features



Overview

- User defined reductions
- Reduction extensions
- Scan operations
- Construct cancellation
- SIMD directives
- Thread affinity
- Masked construct
- Linear clause
- Loop transformations
- Synchronisation hints
- Metadirectives

Not an exhaustive list, nor an in-depth coverage of these topics – the OpenMP spec is now enormous!

Not all compilers may support all these features.

Specification is now very hard to read – often more helpful to consult the examples document at openmp.org

User defined reductions

- Built-in reductions are limited to basic data types (or arrays of these) and a restricted list of operators
 - does now include **min** and **max** in C/C++!
- Use **declare reduction** directive to define new reduction operators
- New operators can then be used in reduction clause.

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```
#pragma omp declare reduction (reduction-identifier :  
typename-list : combiner) [identity(identity-expr)]
```

- **reduction-identifier** gives a name to the operator
 - Can be overloaded for different types
 - Can be redefined in inner scopes
- **typename-list** is a list of types to which it applies
- **combiner** expression specifies how to combine values
- **identity** can specify the identity value of the operator
 - Can be an expression or a brace initializer

Example: merging vectors

```
#pragma omp declare reduction (merge : std::vector<int>
: omp_out.insert(omp_out.end(), omp_in.begin(), omp_in.end()))
```

- Private copies created for a reduction are initialized to the identity that was specified for the operator and type
 - Default identity defined if identity clause not present
- Compiler uses combiner to combine private copies
- **omp_out** refers to private copy that holds combined values
- **omp_in** refers to the other private copy
- Can now use **merge** as a reduction operator.

Reduction extensions

- Can force a reduction to happen at a specified point inside a parallel region with the **scope** construct:

```
#pragma omp parallel shared(a) private(b)
{
    #pragma omp single
    {
        a = 0.0;
    }
    #pragma omp scope reduction(+:a)
    {
        a += foo();
    } // reduction happens here
    b = 2 * a;
}
```

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- Also possible to do reductions on private variables inside a parallel region.
 - each private copy is combined with all the updates to the reduction variable

```
#pragma omp parallel private(a,b)
{
    a = 0.0;
    #pragma omp for reduction(+:a)
    for (i=0; i<N; i++)
        a += foo();
    } // all private copies get updated
    b = 2 * a;
}
```

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Scan operations

- As well as reductions, OpenMP now has support for prefix scan
- For example, as well as computing the total sum with an addition reduction, we can also compute all the partial sums across all the iterations
 - can be inclusive (i.e. k^{th} value includes k^{th} iteration) or exclusive (only values up to iteration $k-1$)

```
!$omp parallel do reduction(inscan,+:x)
do k = 1, n
    x = x + a(k)
!$omp scan inclusive(x)
    b(k) = x
end do
```

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Construct cancellation

- Clean way to signal early termination of an OpenMP construct.
 - one thread signals
 - other threads jump to the end of the construct

```
!$omp cancel construct [if (expr)]  
#pragma omp cancel construct [if (expr)]
```

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where *construct* is **parallel**, **sections**, **do/for** or **taskgroup** cancels the construct

```
!$omp cancellation point construct  
#pragma omp cancellation point construct
```

checks for cancellation (also happens implicitly at cancel directive, barriers etc.)

Example

```
!$omp parallel do private(eureka)
do i=1,n
    eureka = testing(i,...)
!$omp cancel parallel if(eureka)
end do
```

- First thread for which **eureka** is true will cancel the parallel region and exit.
- Other threads exit next time they hit the **cancel** directive

Portable SIMD directives

- Many compilers support SIMD directives to aid vectorisation of loops.
 - compiler can struggle to generate SIMD code without these
- OpenMP 4.0 provides a standardised set
- Use **simd** directive to indicate a loop should be SIMDized

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```
#pragma omp simd [clauses]
```

```
!$omp simd [clauses]
```

- Executes iterations of following loop in SIMD chunks
- Loop is not divided across threads
- SIMD chunk is set of iterations executed concurrently by SIMD lanes

- Clauses control data environment, how loop is partitioned
- **safelen(length)** limits the number of iterations in a SIMD chunk.
- **linear** lists variables with a linear relationship to the iteration space
- **aligned** specifies byte alignments of a list of variables
- **private**, **lastprivate**, **reduction** and **collapse** have usual meanings.
- Also **declare simd** directive to generate SIMDised versions of functions.
- Can be combined with loop constructs (parallelise and SIMDise)

Affinity environment

- Since many systems are now NUMA and SMT, placement of threads on the hardware can have a big effect on performance.
- Increased choices for `OMP_PROC_BIND`
- Can still specify `true` or `false`
- Can now provide a list (possible item values: `master`, `close` or `spread`) to specify how to bind parallel regions at different nesting levels.
- Added `OMP_PLACES` environment variable
- Can specify abstract names including threads, cores and sockets
- Can specify an explicit ordered list of places
- Place numbering is implementation defined
- Good for nested parallel regions
- For MPI + OpenMP needs a mixture of this and batch scheduler to get binding correct

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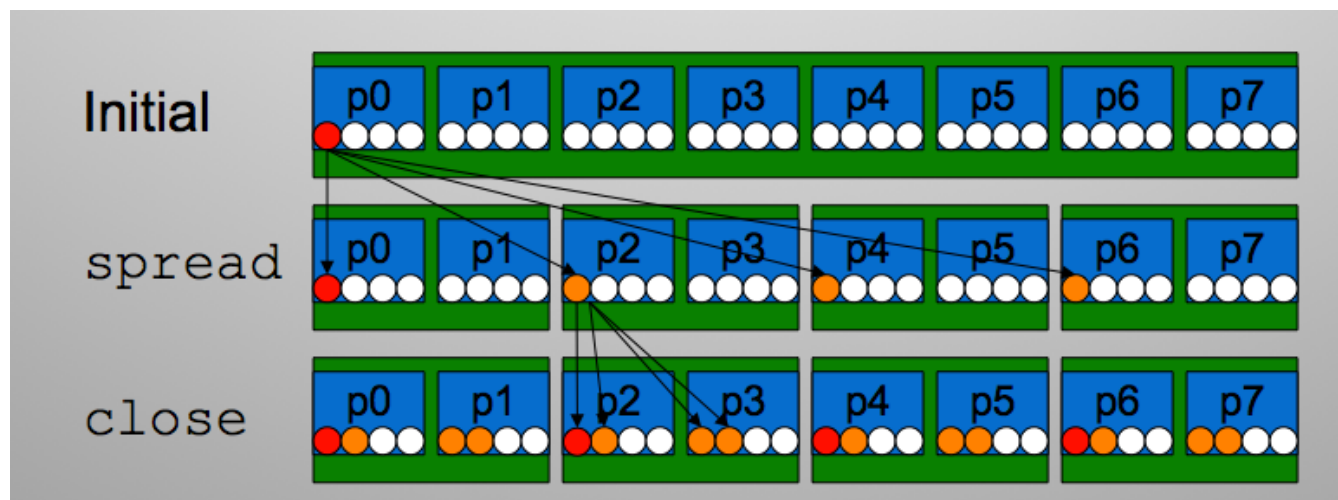
Example

- Two levels of nested parallel regions
- Hardware with 8 cores and 4 hardware threads per core

```
export OMP_NUM_THREADS=4,2
```

```
export OMP_PLACES=threads
```

```
export OMP_PROC_BIND="spread,close"
```



Masked construct

- The **master** construct has been deprecated
- It will still work, but has been renamed **masked**
- Terminology for thread 0 has changed too: *master thread* -> *primary thread*
- Now possible to select threads other than thread 0 to execute the block, with the **filter** clause.
- If the argument to the **filter** clause matches the number of the encountering thread, then that thread will execute the block
 - if more than one thread matches, they will all execute the block

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Masked example

- Execute block of code only on even numbered threads

```
#pragma omp parallel
{
  int includeme;
  if (omp_get_thread_num()%2 == 0) {
    includeme = omp_get_thread_num();
  } else {
    includeme = -1;
  }
  #pragma omp masked filter(includeme)
  {
    // block of code
  }
}
```


Linear clause



- An *induction* variable is a variable in a loop which is incremented by a constant every iteration
- Can prevent straightforward parallelisation of loops
 - refactoring usually trivial, but can be unwanted/tedious
- Can now be handled with **linear** clause on **do/for** construct

```
j=0;  
#pragma omp parallel for linear(j:1)  
for(i=0; i<N; i+=2){  
    b[j] = a[i] * 2.0f;  
    j++;  
}
```

Version 4.5



Loop transformations

- Many compilers support programmer control of loop transformations (e.g. unrolling, tiling, interchanging, reversing) via compiler-specific directives
- OpenMP now supports a portable mechanism for these transformations (and combinations of them).

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```
#pragma unroll full
for(i=0; i<4; i++){
    b[i] = 2.0 * a[i];
}
```

Fully unrolls a sequential loop

```
!$omp parallel do
!$omp tile sizes(5,16)
    do i = 1, 100
        do j = 1, 128
            A(j,i) = j*1000 + i
        end do
    end do
end do
```

Tiles loop nest into 5 x 16 blocks and distributes blocks across threads

Synchronisation hints

- Can give hints about the expected use of synchronisation constructs (critical, atomics, locks) to assist optimisation of these.
- *contended* if we expect threads to wait for each other frequently, or *uncontended* if not.
- *speculative* if using transactional memory might be beneficial, or *nonspeculative* if not.
- **critical** and **atomic** constructs take a **hint** clause, e.g.

```
#pragma omp atomic hint(omp_sync_hint_uncontended)
    a[i]++;
```

- Can initialise locks with **omp_init_lock_with_hint()** instead of **omp_init_lock()**

Monotonic/nonmonotonic schedules



- Older versions of OpenMP mandated that loop chunks are assigned to threads in their logical execution order
 - e.g. for a dynamic schedule, the k^{th} chunk must be assigned to a thread before the $k+1^{\text{th}}$ chunk.
- This restriction has been relaxed: chunks can be assigned in any order
 - can improve data affinity, and reduce synchronisation overheads
- Can enable/disable monotonicity if you want to, e.g.

```
!$omp parallel do schedule(monotonic:dynamic)
```

```
!$omp parallel do schedule(nonmonotonic:dynamic)
```

- Default is monotonic for **static** schedule kind, nonmonotonic otherwise.

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Metadirectives



- Syntax that supports runtime choice of OpenMP directives without having to explicitly write conditional code.
- Can do tests based on program variables, or on the type of offloading device, or on the ISA of the CPU.
- Can also create code variants and select which is used.

```
#pragma omp metadirective \  
when(user={condition(unbal)}): \  
for schedule(guided)) \  
otherwise(for schedule(static))  
for (i=0; i<N; i++){  
    ...  
}
```

```
!$omp metadirective &  
!$omp when(user={condition(unbal)}):&  
!$omp do schedule(guided)) &  
!$omp otherwise(do schedule(static))  
do i=1,N  
    ...  
end do
```

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